KARATINA UNIVERSITY
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BIT 211: COMPUTER SYSTEMS ORGANIZATION

ASSIGNMENT1 (Group Work) **DUE:** 8th November 2024

Instructions: In your groups, attempt all questions.

1) Using a valid example, explain the term Instruction Set Architecture. **(3 Marks)**

2) Some CPUs perform better than other CPUs. Apart from the amount of cache in a CPU, explain THREE other factors that influence the performance of a CPU. **(3 Marks)**

3) For each of the factors in Question (2) above, explain how the factor/characteristic you listed can improve performance. **(3 Marks)**

4) Differentiate between the L1, L2 AND L3 cache memories. **(6 Marks)**

5) Discuss any THREE cache mapping techniques. **(6 Marks)**

6) Your friend has bought a disk that has six platters, 1024 bytes per sector, 12,000 tracks per surface, and an average of 460 sectors per track

   Required: Calculate the capacity of your friend's disk. Show your workings. **(5 Marks)**

7) Show how -65_{10} is represented in the following notations:
   a) Two's complement notation **(2 Marks)**
   b) Signed magnitude notation **(2 Marks)**
   c) Excess notation **(2 Marks)**

8) Explain the role played by the below components of the Von-Neumann architecture:
   a) The Memory Address Register (MAR) **(2 Marks)**
   b) The Memory Data Register (MDR) **(2 Marks)**
   c) The Accumulator **(2 Marks)**
   d) The Program Counter (PC) **(2 Marks)**